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Statistical Methodology for In-House A/B Testing

Narrative Methods Description (Game Live-Ops Context)

1. Purpose and scope

A core part of live operations at a game studio is designing and running controlled experiments — the simplest form being A/B tests — to optimize online products. With dozens of experiments run each year, a coherent, auditable statistical story matters for both product decisions and long-term learning. This note describes how user-level data flow into summaries, how variance reduction and relative effects are defined, how Bayesian inference sits on top of those summaries, and where frequentist diagnostics and known limitations fit. The goal is a single narrative rather than a list of disconnected formulas.

2. Notation

Symbols are introduced once here and reused throughout. We distinguish *population* quantities (unobserved, no hat) from *sample-based estimators* (hat or bar).

Symbol	Meaning
g	Assignment group index; $g \in \{C, 1, 2, \dots, K - 1\}$, with C the control group and $t \in \{1, \dots, K - 1\}$ the t -th treatment variant.
K	Number of groups (control + treatment variants).
$i \mapsto g$	“User i is assigned to group g ” (reads as “ i maps to g ”).
$Y_{i,g}$	User-level outcome (KPI) for user i in group g , after KPI-specific filters.
$\mathbf{x}_i$	Vector of pre-experiment covariates for user i (e.g. pre-period spend, sessions, tenure), with missingness indicators appended.

Symbol	Meaning
n_g	Number of users in group g with a non-null outcome.
μ_g	Population mean of Y in group g , i.e. $\mathbb{E}[Y_{i,g}]$. Estimated by $\hat{\mu}_g$.
$\bar{Y}_g$	Sample mean of Y in group g (an estimator of μ_g).
$\frac{s_g}{\text{SE}(\bar{Y}_g)}$	Sample standard deviation of Y in group g . Estimated standard error of $\bar{Y}_g$.
Δ_t	Population absolute treatment effect of variant t vs. control: $\Delta_t = \mu_t - \mu_C$.
$\hat{\Delta}_t$	Estimator of Δ_t : $\hat{\Delta}_t = \bar{Y}_t - \bar{Y}_C$.
μ_C^{raw}	Control-group population mean on the <i>raw</i> (non-CUPED) KPI. Estimated by $\bar{Y}_C^{\text{raw}}$.
ρ_t	Population relative treatment effect, $\rho_t = \Delta_t / \mu_C^{\text{raw}}$. This is also the latent parameter updated in the Bayesian step (Section 6); no separate symbol is used.
$\hat{\rho}_t$	Sample estimator of ρ_t : $\hat{\rho}_t = \hat{\Delta}_t / \bar{Y}_C^{\text{raw}}$. Treated in the Bayesian step as a noisy measurement of ρ_t .
$\hat{\sigma}_{\text{obs},t}$	Estimated standard error of $\hat{\rho}_t$ from the delta method.
μ_0, σ_0^2	Prior mean and variance for ρ_t .
$\mu_{\text{post},t}, \sigma_{\text{post},t}^2$	Posterior mean and variance of ρ_t given data.
$\tau_0, \tau_{\text{data},t}, \tau_{\text{post},t}$	Precisions: $\tau_0 = 1/\sigma_0^2$, $\tau_{\text{data},t} = 1/\hat{\sigma}_{\text{obs},t}^2$, $\tau_{\text{post},t} = \tau_0 + \tau_{\text{data},t}$.
$\Phi(\cdot)$	CDF of the standard normal $\mathcal{N}(0,1)$ distribution.

3. From randomization to group-level summaries

3.1 Variants, control, and estimands

Users are assigned to a control group and one or more treatment variants indexed by $t \in \{1, \dots, K-1\}$ (non-control test groups). For a fixed key performance indicator (KPI), the outcome $Y_{i,g}$ is observed for each user in the analysis window. Under standard assumptions — stable unit treatment values and no interference between users — contrasts $\mu_t - \mu_C$ identify the average treatment effect of variant t on the user-level outcome. The estimand of primary interest is the *relative* version (for $\mu_C^{\text{raw}} \neq 0$) defined in Section 4.2.

3.2 Means and standard errors at the group level

After preprocessing, analysis works with aggregates by assignment group. Let $\mathcal{J}_g = \{i : i \mapsto g\}$ be the index set of users assigned to group g . Then

$$\hat{\mu}_g = \bar{Y}_g = \frac{1}{n_g} \sum_{i \in \mathcal{J}_g} Y_i, \quad (1)$$

$$s_g^2 = \frac{1}{n_g - 1} \sum_{i \in \mathcal{J}_g} (Y_i - \bar{Y}_g)^2, \quad (2)$$

$$\widehat{\text{SE}}(\bar{Y}_g) = \frac{s_g}{\sqrt{n_g}}. \quad (3)$$

Each assignment group contributes a pair $(\bar{Y}_g, \widehat{\text{SE}}(\bar{Y}_g))$ that summarizes the evidence for that variant or the control.

4. Absolute and relative treatment effects

4.1 Absolute uplift

For each treatment variant $t \in \{1, \dots, K - 1\}$, the absolute uplift estimator is

$$\hat{\Delta}_t = \bar{Y}_t - \bar{Y}_C. \quad (4)$$

Randomization makes $\bar{Y}_t$ and $\bar{Y}_C$ approximately independent (conditional on the realized sample sizes n_t, n_C), so

$$\widehat{\text{SE}}(\hat{\Delta}_t) = \sqrt{\widehat{\text{SE}}(\bar{Y}_t)^2 + \widehat{\text{SE}}(\bar{Y}_C)^2}. \quad (5)$$

Equations (4)–(5) support standard large-sample normal reporting for $\hat{\Delta}_t$ and feed the relative-uplift step.

4.2 Relative uplift and a consistent denominator

Stakeholders typically interpret results as percentage change relative to control. Two conventions matter here. First, the denominator is *always* the control mean on the **raw** (non-CUPED) KPI, $\bar{Y}_C^{\text{raw}}$, even when the numerator $\hat{\Delta}_t$ uses a CUPED-adjusted outcome (Section 5). Second, only the numerator carries the variance-reduction benefit; the business denominator is fixed to the raw baseline so that “% lift” is interpretable on a single scale across experiments. The relative uplift estimator is

$$\hat{\rho}_t = \frac{\hat{\Delta}_t}{\bar{Y}_C^{\text{raw}}}. \quad (6)$$

4.3 Uncertainty for ratios via the delta method

The relative uplift is a nonlinear function of two estimated means. Write $\hat{\rho}_t = f(\hat{\Delta}_t, \bar{Y}_C^{\text{raw}})$ with $f(a, b) = a/b$. A first-order Taylor expansion of f around the population values $(\Delta_t, \mu_C^{\text{raw}})$ gives

$$\hat{\rho}_t \approx \frac{\Delta_t}{\mu_C^{\text{raw}}} + \frac{1}{\mu_C^{\text{raw}}}(\hat{\Delta}_t - \Delta_t) - \frac{\Delta_t}{(\mu_C^{\text{raw}})^2}(\bar{Y}_C^{\text{raw}} - \mu_C^{\text{raw}}), \quad (7)$$

so the coefficients in the delta-method variance are exactly the two partial derivatives of the ratio function. Assuming $\hat{\Delta}_t$ and $\bar{Y}_C^{\text{raw}}$ are approximately independent (a standard working assumption under randomization),

$$\widehat{\text{Var}}(\hat{\rho}_t) \approx \left(\frac{1}{\bar{Y}_C^{\text{raw}}} \right)^2 \widehat{\text{Var}}(\hat{\Delta}_t) + \left(\frac{\hat{\Delta}_t}{(\bar{Y}_C^{\text{raw}})^2} \right)^2 \widehat{\text{Var}}(\bar{Y}_C^{\text{raw}}), \quad (8)$$

where $\widehat{\text{Var}}(\hat{\Delta}_t) = \widehat{\text{SE}}(\hat{\Delta}_t)^2$ and $\widehat{\text{Var}}(\bar{Y}_C^{\text{raw}}) = \widehat{\text{SE}}(\bar{Y}_C^{\text{raw}})^2$. We report

$$\hat{\sigma}_{\text{obs},t} = \sqrt{\widehat{\text{Var}}(\hat{\rho}_t)}. \quad (9)$$

The same gradient logic extends to KPIs defined as ratios of two per-user metrics and to compound relative uplifts built from numerator and denominator KPIs, yielding a single scalar $\hat{\sigma}_{\text{obs},t}$ in each case.

5. CUPED as a variance-reduction layer before inference

Outcomes in games are often predictable from pre-experiment engagement or spend. **Controlled-experiment Using Pre-Experiment Data (CUPED)** exploits this by regressing the in-period outcome on pre-period covariates (with missingness indicators) at the user level,

$$Y_i = \beta_0 + \mathbf{x}_i^\top \beta + \varepsilon_i, \quad (10)$$

where:

- β_0 is the intercept (the expected outcome for a user whose covariates are all at the reference level / zero).
- β is the vector of covariate slopes; β_j is the expected change in Y_i per unit change in the j -th pre-experiment covariate, holding the others fixed.
- $\mathbf{x}_i$ is the pre-experiment covariate vector for user i ; each component is observed *before* randomization and therefore cannot have been affected by treatment.
- ε_i is the residual noise (what the covariates cannot explain), assumed to have zero conditional mean.

Ordinary least squares produces estimates $\hat{\beta}_0$ and $\hat{\beta}$, and the fitted value for user i is

$$\hat{Y}_i = \hat{\beta}_0 + \mathbf{x}_i^\top \hat{\beta}. \quad (11)$$

Intuitively, $\hat{Y}_i$ is the “best guess” of user i ’s in-period outcome made using only pre-experiment information. CUPED then replaces the raw outcome by the residual — the part of Y_i that the pre-period covariates could not predict —

$$Y_i^{\text{CUPED}} = Y_i - \hat{Y}_i. \quad (12)$$

Because $\hat{Y}_i$ is constructed from pre-randomization variables, subtracting it removes noise without distorting the treatment effect; in expectation, $\mathbb{E}[Y_t^{\text{CUPED}} - Y_C^{\text{CUPED}}] = \Delta_t$. Group means and standard errors (1)–(3) are then recomputed with Y_i^{CUPED} in place of Y_i , yielding smaller s_g and hence smaller $\widehat{\text{SE}}(\bar{Y}_g)$. Only after this step does the pipeline form $\hat{\Delta}_t$, $\hat{\rho}_t$, and $\hat{\sigma}_{\text{obs},t}$: variance reduction is part of the measurement model, not an afterthought.

6. Bayesian inference on relative uplift

6.1 Normal–normal conjugate update

The parameter we want to learn in this step is the same ρ_t introduced in Section 4 — the population relative treatment effect of variant t vs. control. Its sample estimator $\hat{\rho}_t$, with standard error $\hat{\sigma}_{\text{obs},t}$ from (9), is treated as a noisy measurement of this unknown ρ_t . We do *not* introduce a separate Greek letter (such as θ) for the latent parameter; the population vs. estimator distinction is carried entirely by the hat, so ρ_t is what we want to know and $\hat{\rho}_t$ is what the data tell us. The prior–likelihood pair is:

$$\rho_t \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad \hat{\rho}_t | \rho_t \sim \mathcal{N}(\rho_t, \hat{\sigma}_{\text{obs},t}^2). \quad (13)$$

The normal prior and normal sampling model are conjugate. With precisions $\tau_0 = 1/\sigma_0^2$ and $\tau_{\text{data},t} = 1/\hat{\sigma}_{\text{obs},t}^2$, the posterior is $\rho_t | \hat{\rho}_t \sim \mathcal{N}(\mu_{\text{post},t}, \sigma_{\text{post},t}^2)$ with

$$\tau_{\text{post},t} = \tau_0 + \tau_{\text{data},t}, \quad (14)$$

$$\mu_{\text{post},t} = \tau_{\text{post},t}^{-1} (\tau_0 \mu_0 + \tau_{\text{data},t} \hat{\rho}_t), \quad (15)$$

$$\sigma_{\text{post},t}^2 = \tau_{\text{post},t}^{-1}. \quad (16)$$

The posterior mean (15) is a precision-weighted blend of prior belief μ_0 and the data $\hat{\rho}_t$; the prior variance σ_0^2 controls how much the posterior shrinks toward μ_0 .

6.2 What is not modeled jointly

Each treatment variant receives its own marginal posterior $p(\rho_t | \text{data})$. There is no joint model over all variants that would fully capture dependence across multiple test groups; multi-variant statements rely on marginal posteriors and simple decision rules below, not on a single multi-test hierarchical decision problem across variants.

6.3 Decision-oriented summaries

- **Chance to beat control:** $P(\rho_t > 0 | \text{data}) = 1 - \Phi(-\mu_{\text{post},t}/\sigma_{\text{post},t})$, the posterior probability that the true relative uplift is positive.
- **Credible intervals:** equal-tailed intervals from the posterior quantile function, e.g. $\mu_{\text{post},t} \pm 1.96 \sigma_{\text{post},t}$ for a 95% interval.

A simple winning rule selects the variant with largest posterior mean unless all favor the control; “inverse” KPIs (lower is better) use the analogous minimum rule.

7. Frequentist reference and design diagnostics

Alongside Bayesian summaries, the workflow retains frequentist objects — group means, standard errors, and period splits — as a transparent view of raw and adjusted series across pre-test, in-test, and post-test windows. This dual view helps stakeholders who think in classical terms while the primary decision language remains posterior-based for lift.

7.1 Sample ratio mismatch (assignment balance)

Before trusting contrasts, check whether realized assignment counts n_g match the intended allocation $\{np_g\}_{g=1}^K$ (with $\sum_g p_g = 1$; under equal allocation $p_g = 1/K$). The Pearson chi-square goodness-of-fit statistic is

$$X^2 = \sum_{g=1}^K \frac{(n_g - np_g)^2}{np_g}, \quad n = \sum_{g=1}^K n_g, \quad (17)$$

where H_0 is that the realized split matches the intended one, and $X^2 \overset{\text{approx.}}{\sim} \chi_{K-1}^2$ under H_0 . A small p -value flags **sample ratio mismatch**: possible issues with randomization, eligibility filters, or tracking, rather than a treatment effect on the KPI.

8. Worked example: a battle-pass offer UI test

To make the notation above concrete, we walk a small example end-to-end using pseudo data. All numbers are invented for illustration and are rounded for readability; the arithmetic is consistent with the formulas in Sections 3–7.1.

Setting. A mobile RPG studio tests two redesigns of the battle-pass purchase flow. Users are randomized on login into three groups ($K = 3$):

- $g = C$: existing flow (control);
- $g = 1$: new offer UI (treatment variant 1);
- $g = 2$: new offer UI with an aggressive price anchor (treatment variant 2).

The primary KPI is 14-day per-user revenue (ARPU, in USD). The pre-experiment covariate vector $\mathbf{x}_i$ contains 14-day pre-period ARPU, 14-day session count, account tenure in days, and corresponding missingness indicators ($\mathbf{x}_i \in \mathbb{R}^6$).

Assignment-balance check (Section 7.1). The intended split is $p_C = p_1 = p_2 = 1/3$. Realized counts are

Group	n_g
C	5{,}024
1	5{,}012
2	4{,}964
Total n	15{,}000

Applying (17) with $np_g = 5,000$,

$$X^2 = \frac{(24)^2 + (12)^2 + (-36)^2}{5,000} = \frac{576 + 144 + 1,296}{5,000} \approx 0.40.$$

With $K - 1 = 2$ degrees of freedom, $p \approx 0.82$, so the assignment balance check passes and we proceed.

Raw group summaries (Section 3.2). Suppose the raw in-period ARPU statistics are:

Group	$\bar{Y}_g$ (USD)	s_g (USD)	$\widehat{\text{SE}}(\bar{Y}_g)$ (USD)
C	5.00	14.0	0.197
1	5.40	14.1	0.199
2	5.15	14.0	0.199

Each SE is $s_g/\sqrt{n_g}$ from (3); e.g. $14.0/\sqrt{5,024} \approx 0.197$. In particular, $\bar{Y}_C^{\text{raw}} = 5.00$ (this value is fixed as the relative-uplift denominator throughout).

CUPED step (Section 5). We fit (10) on the pooled sample. Suppose the OLS fit yields (rounded)

$$\hat{\beta}_0 \approx 1.20, \quad \hat{\beta}_{\text{pre-ARPU}} \approx 0.65, \quad \hat{\beta}_{\text{sessions}} \approx 0.030, \quad \hat{\beta}_{\text{tenure}} \approx 0.0025,$$

where the three leading slopes attach to pre-period ARPU, pre-period session count, and account tenure (the missingness-indicator coefficients are omitted for brevity). A user with pre-period ARPU = \$3.00, 20 sessions, and tenure 200 days has fitted value

$$\hat{Y}_i \approx 1.20 + 0.65 \cdot 3.00 + 0.030 \cdot 20 + 0.0025 \cdot 200 \approx 4.25,$$

and residual $Y_i^{\text{CUPED}} = Y_i - \hat{Y}_i$. Across all users the residuals have sample SD roughly 50% smaller than the raw outcome; assume post-CUPED $s_g^{\text{CUPED}} \approx 7.0$ in each group, giving

$$\widehat{\text{SE}}(\bar{Y}_g^{\text{CUPED}}) \approx \frac{7.0}{\sqrt{5,000}} \approx 0.099 \quad \text{for } g \in \{C, 1, 2\}.$$

Absolute and relative uplift (Section 4). The CUPED residual differences are (for illustration, unchanged in magnitude from the raw differences but with much tighter SEs):

$$\hat{\Delta}_1 = 0.40, \quad \widehat{\text{SE}}(\hat{\Delta}_1) = \sqrt{0.099^2 + 0.099^2} \approx 0.140,$$

9. Tensions with current practice and directions for improvement

The methodology above matches a practical pipeline: CUPED for measurement, delta-method uncertainty, then conjugate Bayes on relative uplift with an informed prior. In parallel, the following issues from ongoing practice are worth stating explicitly for peer or course review.

Rigid informed priors. A common implementation choice is to encode prior beliefs about relative lift through fixed hyperparameters (for example, $\mu_0 = 0.05$ with $\sigma_0 = 0.05$ as in Section 8). Such a one-size-fits-all prior can be problematic: it may reduce effective power for true effects smaller than μ_0 , while strongly shrinking estimates when the true lift is much larger, which can understate the full business impact of a large win. A natural improvement is to replace a hard-coded (μ_0, σ_0) with **dynamic priors** informed by historical experiments, segment-level baselines, and analyst input, while documenting sensitivity of conclusions to those choices.

Subgroup analyses and multiple testing. Teams often slice results by platform, geography, or player tier to hunt for uplift in subsets. Even when the overall experiment is well powered, exploratory subgroup comparisons multiply the number of hypotheses and inflate the risk of Type I error (false positives): a null overall result can still produce a bright-looking segment by chance. Mitigations used in the wider literature include Bonferroni-style correction across a prespecified family of tests, hierarchical or partial pooling across segments, or preregistering a small number of subgroup hypotheses. The core platform narrative above does not, by itself, enforce such corrections; any improvement plan should separate **confirmatory** (prespecified) from **exploratory** slicing and apply error-rate control where appropriate.

10. Closing the loop: end-to-end logic

In summary, the methodology chains four ideas:

1. **Randomized assignment** identifies contrasts $\mu_t - \mu_C$ between the control group and each treatment variant at the user level.
2. **CUPED** (equations (10)–(12)) removes predictable variation so that group-level means carry more information per user.
3. **Delta-method propagation** (equations (7)–(9)) turns those means into a relative uplift $\hat{\rho}_t$ and a standard error $\hat{\sigma}_{\text{obs},t}$ on the business scale.
4. **Bayesian updating** (equations (13)–(16)) merges $\hat{\rho}_t$ with an informed prior on ρ_t , yielding posterior probabilities and credible intervals aligned with product language.

Assignment-balance checks (17) and frequentist tabular views sit alongside this story: the former guards the design; the latter keeps raw evidence visible while priors and posteriors encode judgment and uncertainty about lift.

Topics for external review. Worth scrutinizing are: (i) treating $\hat{\sigma}_{\text{obs},t}$ as known rather than propagating full uncertainty from CUPED and the delta method; (ii) independence assumptions in ratio metrics; (iii) marginal posteriors across variants and KPIs versus joint decision rules; (iv) clustering or network effects not captured by group-level standard errors; and (v) the prior and multiple-testing points raised above.